

Initial Analysis of the Banknote Authentication Dataset

Introduction

The UCI Machine Learning Repository provides the Banknote Authentication dataset, which has two classes (Genuine or Fake) and four characteristics. An 80-20 train-test split with random shuffling was used, and the dataset was pre-processed without normalisation. A neural network of two output neurons, four input neurons, and one hidden layer was used. Weights were initialised at random, and the sigmoid activation function was used.

Methodology

Different hidden layer sizes (10, 50, 100, and 150 neurons) and learning rates (0.1, 0.3, and 0.5) were used to train the neural network. Mean Squared Error (MSE) was used as the loss function for the 500 epochs of training. Training time, final accuracy, and test accuracy across epochs were used to evaluate performance.

Results

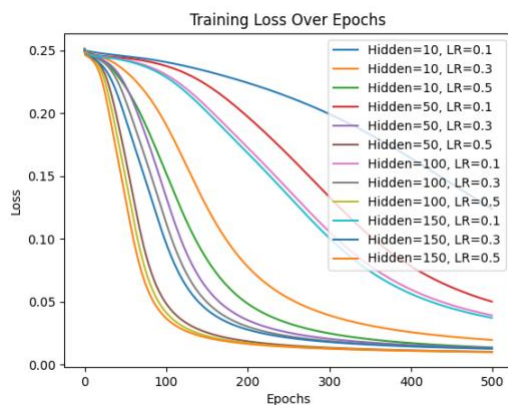


Figure 1: graph showing training loss over Epochs.

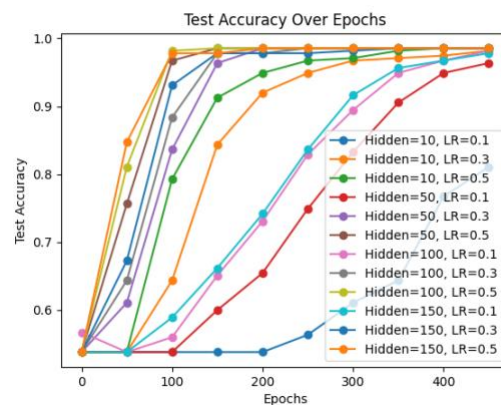


Figure 2: graph showing test accuracy over Epochs.

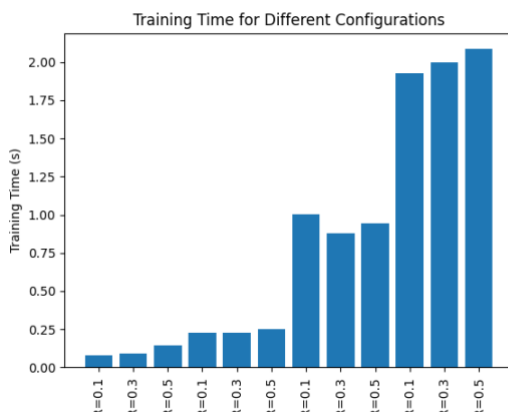


Figure 3: graph showing training time for different configurations.

- Larger hidden layers and greater learning rates result in faster convergence, shown in figure 1, which shows the training loss decreasing over epochs for all configurations.
- Figure 2 shows how test accuracy increased over time, with higher accuracy achieved by configurations with more neurons and faster learning rates.

- Figure 3 displays how the training time increased as the hidden layer's neuron count increased. Training times were significantly greater for configurations with 150 neurons compared the ones with 10.

Comparative Analysis

- Python implementations have shown to obtain faster training times compared to the results obtained in MATLAB, especially for larger networks.
- Both MATLAB and Python achieved around similar final accuracies, however Python showed more accurate results with convergence speed.